University of Notre Dame

Department of Computer Science and Engineering

CSE 40567/60567: Computer Security

Assignment Date:	2/26/26
Submission Date:	
Student Name:	Mwayi Kashoko
Challenge of HW difficulty: (1 – least challenging to 5 – most challenging) 4, Vigenere cipher was the most challenging however mostly this homework was just time consuming.	

* This is the HW2 assignment which contains 100 points. You are encouraged to finish independently. Any submitted work that is copied from any source or too similar to be an independent write-up will not be given credit. Please submit this homework (as a word or PDF file including the cover page) to ND Canvas by **11:59pm on Thursday, 02/26/2026.**

Problem 1 (5 pts). Rail-Fence Cipher is a type of transposition cipher. Please decrypt the cyphertext "**CWEODITRLN**" with 3 "rails".

Note: In the Rail-Fence cipher, the plaintext is written downwards diagonally on successive "rails" of an imaginary fence, then moving up when the bottom rail is reached, down again when the top rail is reached, and so on until the whole plaintext is written out. The ciphertext is then read off in rows. For your reference, the right figure shows an example of how to use 3 "rails" for encryption.

"WE ARE DISCOVERED. RUN AT ONCE."

with 3 "rails",

```
W . . . E . . . C . . . R . . . U . . . O . . .
. E . R . D . S . O . E . E . R . N . T . N . E
. . A . . . I . . . V . . . D . . . A . . . C .
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COLD WINTER

Problem 2 (10 pts). Please crack the Caesar cipher "**JVTWBALYZJPLUJL**" using statistical analysis. You are asked to find the key and use it for decryption (5 pts). Please also show your steps (5 pts).

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Shifted by 7

Problem 3 (25 pts). The following message is a Vigenère cipher.

```
TTEUM GQNDV EOIOL EDIRE MQTGS DAFDR CDYOX IZGZP PTAAI TUCSI XFBXY SUNFE
SQRHI SAFHR TQRVS VQNBE EEAQG IBHDV SNARI DANSL EXESX EDSNJ AWEXA ODDHX
EYPKS YEAES RYOET OXYZP PTAAI TUCRY BETHX UFINR
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- Please find the key (5 pts) and decipher it (5 pts).
- Show your steps in detail (15 pts): i.e., using the method introduced in the class to break the cipher (references: "Lecture_CSE 40567_Spring 2026_W4_L05_Cryptography (Caesar & Vigenère Ciphers).pdf" and Assigned Reading "AR-4_CSE 40567_W4_T2-Cryptography (Caesar & Vigenère Ciphers).pdf"). More specifically, you need to show the repetitions you find (5 pts), how you establish the period and exploit index of coincidence to facilitate your estimation (5 pts), how you solve each part after you break the message into *n* parts – i.e., *n* denotes the established period (5 pts). You may write code to help solve the problem.

Note that only providing the key and decrypted text won't get full credits – i.e., you will need to show the key steps as stated above to illustrate how you solve the problem.

Letters	Start	End	Gap Length	Factors of Gap Length
DV	8	88	80	1,2,4,5,8,10,16,20,40,80
QN	6	76	70	1,2,5,7,10,14,35,70
DA	25	95	70	1,2,5,7,10,14,35,70
PPTAAITUC	39	139	100	1,2,4,5,10,1020,25,50,100
ED	15	105	90	1,2,3,5,6,9,10,15,18,30,45,90
ES	59	128	69	1,3,23,69
SN	90	107	17	1,17

Key: AMAZE
Text without formatting: THEVI GENER ECIPH ERISA METHO DOFEN CRYPT INGAL PHABE TICTE XTBYU SINGA SERIE SOFIN TERWO VENCA ESARC IPHER SBASE DONTH ELETT ERSOF AKEYW ORDIT EMPLO YSAFO RMOFP OLYAL PHABE TICSU BSTIT UTION

Decrypted Text: The Vigenere cipher is a method of encrypting alphabetic

text by using a series of interwoven Caesar ciphers based on the letters of a keyword. It employs a form of polyalphabetic substitution.

Problem 4 (10 pts). If one-time pads are provably secure, why are they rarely used in practice?

You would have to deliver the one-time pad to the receiver which is risky. Need a one-time pad for every message sent. The one-time pad has to be the same size as the data encrypted which is not always efficient.

Problem 5 (10 pts). By using RSA algorithm, Bob wants to send Mary message "BRIGHT" (01 17 08 06 07 19) in confidence (confidentiality-assured) and authenticated (integrity-checked). Given the following keys:

– Bob's keys: public (17, 77); private: 53
– Mary's keys: public: (37, 77); private: 13

(1) Please show how Bob will encipher the message.
e=17
n=77
d=53
m_n=[1, 17, 8, 6, 7, 19]
c=m^e mod n

[1, 19, 57, 41, 28, 24]

(2) Please show how Mary will decipher the received message.

e=37
n=77
d=53
m_n = [1, 19, 57, 41, 28, 24]

[1, 17, 8, 6, 7, 19]

code attached in js file sent.

Note: You may leverage PowerMod Calculator to facilitate your calculation:
https://www.boxentriq.com/code-breaking/modular-exponentiation

Problem 6 (40 pts). The following message was enciphered using DES algorithm with the key of "LOVECSND" under electronic code book mode (ECB).

Ciphertext in binary format:
1100101011101101101000100110010101011111101101110011100001110011
Key: "LOVECSND" in binary format:
0100110001001111010101100100010101000011010100110100111001000100

- Please decipher it (10 pts).
- Please show key steps (20 pts). You may take references of "Lecture_CSE 40567_Spring 2026_W5_L06_Cryptography_ DES Algorithm.pdf", "Lecture_CSE 40567_Spring 2026_W5_L07_ Cryptography_ DES Algorithm (Cont).pdf" and Assigned Reading "AR-5_CSE 40567_W5_T2-Cryptography (DES Algorithm Illustration).pdf"). More specifically, you need to show the 16 generated round keys (10 pts), the output of *f* function in each iteration, *LnRn* (1<=n<=16) in each iteration (10 pts).
- Please post your code in GitHub and submit the link of your posted code (10 pts).

Note that only providing the decrypted text won't get full credits – i.e., you will need to show the key steps as stated above to illustrate how you solve the problem.

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All work is shown as code in the GitHub Link Here:
https://github.com/MwayiKashoko/ComputerSecurity/tree/main/Homework2